

Argmax Decoding Causes Liquidity Failure in a Market of Language-Model Traders

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Abstract

We populate a continuous double auction with traders whose decisions are made by a typed language model that returns a probability distribution over actions, and vary only how that distribution is consumed. Taking the model’s modal action—the default in virtually every deployment—makes the market stop trading after shocks: following a positive shock, the market clears in 65.6% of periods against 91.3% after a negative shock, a gap of 25.8 percentage points (95% CI [17.3, 34.2], $t = 6.37$, $n = 20$ preregistered). Sampling from the same distribution reduces the gap to 3.1 points. Two algorithmic baselines that are symmetric by construction show no gap. The mechanism is direct and observable: 100% of non-trading periods under modal decoding had one side of the order book entirely empty, against 3.4% for the zero-intelligence baseline. Because every trader conditions on the same public signal, a deterministic policy makes them act identically, and a market of unanimous buyers has no counterparty. The effect decomposes into two parts: unanimity sets its magnitude, while a standing directional bias in the arm’s order flow determines which shock it strikes.

We test this mechanism with a preregistered prediction it could have failed: if unanimity is the cause, the effect must vanish as the market grows. It does, from 37.0 points at four traders to 4.2 at thirty-two (-0.112 per doubling, $t = -6.91$). A companion prediction—that sampling would show no such decay—was falsified: sampling herds too, seven times more weakly, on the same curve. We therefore report a single mechanism whose strength is set by how sharply the decode rule collapses the model’s distribution, rather than two regimes.

Separately, we find that ordinary domain wording induces a 25-point asymmetry in the model’s stated conviction between buying and selling at equal mispricing, which vanishes under mirror-image phrasing. A preregistered test found this does *not* move pricing error, and we report that null.

All experiments were preregistered, including the two hypotheses that failed.

1 Introduction

A language model asked to choose among actions returns, in principle, a distribution. Almost every deployment discards it and takes the mode. For a single decision this is correct: the mode is the most likely action. This paper asks what happens when many agents driven by the same model make related decisions at the same time.

The answer, in a market, is that the market stops. Modal decoding makes agents conditioning on shared information act identically. In a setting that requires counterparties, unanimity is not consensus; it is the absence of trade. We measure this in a continuous double auction, isolate the mechanism, and confirm it with a prediction that could have failed and did not.

Our contributions:

1. **A liquidity failure caused by decoding, not by reasoning.** The model’s modal action is *correct* on both sides at every mispricing in our static probe grid, and at the shock itself in the market it takes the correct side in both directions. The failure is in the aggregation of decisions that are individually defensible.
2. **A directly observed mechanism, in two parts.** We instrument both sides of the order book and show that under modal decoding, non-trading periods lack a counterparty essentially always, while for symmetric baselines they do not. Unanimity sets the magnitude; a standing directional bias in order flow sets which shock it strikes. Our own first account used only the former and is corrected in Section 6.
3. **A falsifiable size-scaling test.** The mechanism implies the effect must decay with market size at a rate set by the agreement rate. We registered this before running it; it held.
4. **A prompt-sensitivity result, and a null.** Domain-idiomatic wording induces a large conviction asymmetry invisible to anyone reading only the modal action. A preregistered test found it does not move pricing error.

We emphasise the failures. Four preregistered predictions in this project did not survive, and each is reported: the wording-to-prices hypothesis (C1, a clean null), the original mechanism (Section 5), the sampling-amplification prediction, and the claim that sampling would show no size decay (Section 7). The result that survived did so against tests designed to kill it.

2 Related work

Market design and baselines. The experimental market tradition begins with Smith [1962]. Our lower baseline is the zero-intelligence trader of Gode and Sunder [1993], whose result—that budget-constrained random traders achieve high allocative efficiency—makes a random arm a genuine floor rather than a strawman, and motivates the value clamp that prevents any arm in our design from knowingly trading at a loss.

LLM agents in simulated markets. Lopez-Lira [2025] builds a simulated stock market with a persistent order book in which heterogeneous LLM agents submit structured decisions, and reports that agents adhere consistently to assigned strategies while the market reproduces price discovery, bubbles and underreaction. That work also notes that prompts can generate correlated behaviours affecting market stability, which is the phenomenon we isolate and measure here.

Correlation as systemic risk. Closest to our result is Ross et al. [2026], who argue that shared training and architecture make more capable models behave more similarly, creating a non-diversifiable risk floor, and who test this with LLM traders of varying capability in an agent-based market. They find correlated behaviour that *increases* with capability, and that it becomes a liability when agents share a misinformation environment.

Our contribution is complementary and, we think, orthogonal in a useful way. They vary the model and attribute correlation to properties of the model—its training, its architecture, its capability. We hold the model *fixed* and vary only how its output distribution is consumed. The

correlation we measure is therefore not a property of the model at all but of a deployment choice, and it is one a practitioner controls directly: our argmax and sampling arms issue identical requests and read identical responses, differing only in a single line of client code. Where Ross et al. [2026] identify a risk floor set by model similarity, we show a large component of realised correlation sitting above that floor, removable without changing models.

Herding and cascades. The classical account of correlated action is the informational cascade of Bikhchandani et al. [1992], in which an agent rationally ignores private information after observing enough predecessors. Our mechanism is deliberately simpler and should not be confused with it. Agents in our market do not observe one another’s decisions and draw no inference from them; they condition on a common public signal and a deterministic policy maps that signal to the same action. There is no learning, no sequence, and no inference—only a degenerate policy. The resulting unanimity looks like herding and has the same market consequence, but it arises without any of the cascade machinery, which is why it can be switched off at the decoder.

Decoding and diversity. That greedy decoding collapses output diversity is long established in text generation: Holtzman et al. [2020] show that maximisation-based decoding yields repetitive, degenerate text and introduce nucleus sampling as a remedy. The quality–diversity tension is well understood *within* a single generation. We report a consequence one level up: when many agents share a decoder, the diversity that sampling preserves is not a stylistic property of one output but the raw material a market needs to clear. Sampling is individually worse and collectively necessary.

Confidence and calibration. Kadavath et al. [2022] study whether language models can predict the correctness of their own answers, and subsequent work has documented systematic overconfidence in verbalised confidence. Our Experiment 5 touches this literature from an unusual angle: the model we use returns a probability distribution natively rather than a verbalised estimate, and we find that ordinary domain wording shifts that distribution by 25 points without changing the selected option. Any calibration measurement taken over such a model inherits whatever asymmetry the question’s phrasing introduced.

What we could not find. We did not find prior work isolating the decode rule as a cause of market-level liquidity failure, nor a preregistered size-scaling test of a herding mechanism in an LLM-populated market. We would welcome pointers; this survey was conducted by the authors and is not exhaustive.

3 The instrument

3.1 Market

A continuous double auction in one abstract good. Prices are integer ticks; matching is price-then-time priority; an incoming order executes at the resting order’s price. A trader’s resting orders are withdrawn at the top of its turn, so each trader has at most one live order. Submitting on one side cancels the trader’s resting orders on the other, preventing wash trades without leaving a crossed book.

Accounts are conserved by construction. Each trade moves cash from buyer to seller and inventory from seller to buyer, one for one. Short sales and borrowing are disallowed, and every order is pre-checked against uncommitted cash or uncommitted inventory. A property test submits 20,000 random orders and asserts after every one that total cash and total units are exactly their endowed values, that no account is negative, that no trader has committed more than it holds, and that the book is never crossed.

Order size is fixed at one unit, so aggressiveness is the only lever an arm has; position sizing is a second channel and is deliberately disabled.

3.2 Fundamental and information

The fundamental F_t is piecewise constant. For the size and liquidity experiments we use a *matched-jump* process: jumps are evenly spaced, of identical magnitude, strictly alternating in direction, and even in number, so that every up window is paired with a down window of the same size between the same two price levels. This makes the up-versus-down comparison within-run and matched, which matters: under a random-jump process the per-run noise on this statistic has standard deviation 0.74, large enough that a single seed of the symmetric baseline returned a spurious -0.82 gap.

Each trader’s private value is $F_t + \varepsilon_{i,t}$ with $\varepsilon \sim \mathcal{N}(0, 5)$, redrawn each period. All experiments here use full information; the delayed and noisy signal treatments are implemented but not reported.

3.3 The decision contract

Code owns matching, budgets, order sizes and prices. An agent answers three typed questions and nothing else: direction (**buy/sell/pass**), urgency on a five-level scale, and whether the signal is already reflected in the book. A code-owned rule maps urgency to an integer tick, anchored on the book and hard-clamped at the agent’s own private value.

That clamp is load-bearing. *An agent cannot express a losing price*, because it never expresses a price. A 20,000-case property test asserts that no quote is ever on the wrong side of the quoting agent’s value. Consequently every effect we report is about direction and timing, never about arithmetic.

The question wording is frozen and hash-locked; a test fails if any instruction or criterion changes. This is the defence against unrecorded prompt iteration.

3.4 Arms

All four arms emit the same object, including a distribution over **{buy, sell, pass}**, which makes them comparable.

- **ZI**: random direction and urgency, fixed policy distribution, value-clamped so it never knowingly trades at a loss.
- **NBR**: logit best-response over the same urgency grid, scored by expected surplus times a fill probability. Exactly symmetric between sides by construction; a buy at mispricing $+x$ and a sell at $-x$ receive identical probability.
- **Jev-argmax**: the model’s modal action.

- **Jev-sample:** a draw from the model’s distribution.

The two model arms read one response, so for a given observation they are one API call and differ only in the decode rule.

4 The model and its measurement properties

The model is a commercially available typed-decision model that returns, for a multiple-choice question, a selected option together with a probability over all options. Three properties affect measurement and are reported because they constrain what can be claimed.

Non-determinism. Identical requests return different answers. Four repetitions of one request returned urgency scores of 2.84, 2.79, 2.82 and 2.74 and buy probabilities of 0.98, 0.99, 0.99, 0.99. There is no seed parameter. Run-to-run noise is roughly ± 0.05 on a 0–4 urgency scale and ± 0.01 on probabilities. *Reproducibility therefore requires the response archive, not merely the code:* a rerun without it does not reproduce. All 110,805 responses are retained and every reported figure is regenerable from them.

Quantisation. Probabilities are returned to two decimal places, so three-option distributions routinely sum to 0.99 or 1.01. We accept and renormalise within tolerance.

Undocumented failure modes. The published status codes are incomplete; the service returns bare 5xx from its edge under load. A client written to the documentation alone will treat these as fatal.

5 Experiment 1: wording, conviction, and a null

5.1 Domain wording induces a conviction asymmetry

Our first criteria were written in ordinary domain language: *buy* as “the good is underpriced. . . you want to acquire a unit”, *sell* as “the good is overpriced. . . you want to dispose of a unit”. At equal absolute mispricing the model placed 0.988 on the profitable side when buying was correct and 0.738 when selling was correct, a gap of 0.250, with the missing mass on *pass*.

The gap survives a second price level (+0.225), reversing the option order (+0.235), and renaming the options neutrally (+0.185). It does not survive rewriting the two criteria as exact structural mirrors—“value is *above* the price” versus “*below* the price”—under neutral instructions, where the gap is +0.005 (Figure 1).

Two points matter for practice. First, the modal action is *correct on both sides at every point tested*: a practitioner reading only the selected option sees nothing wrong. Second, the wording that induces the skew is the natural one; symmetry had to be engineered deliberately.

5.2 It does not move pricing error

We predicted this skew would slow price discovery after negative shocks. A four-seed exploratory run supported it (+0.63, four of four seeds positive, Holm $p = 0.061$). We registered the test and ran it on twenty fresh seeds. It failed: -0.184 , 95% CI $[-0.479, +0.111]$, seven of nineteen seeds positive, one-sided $p = 0.897$. The confirmatory interval excludes the exploratory estimate.

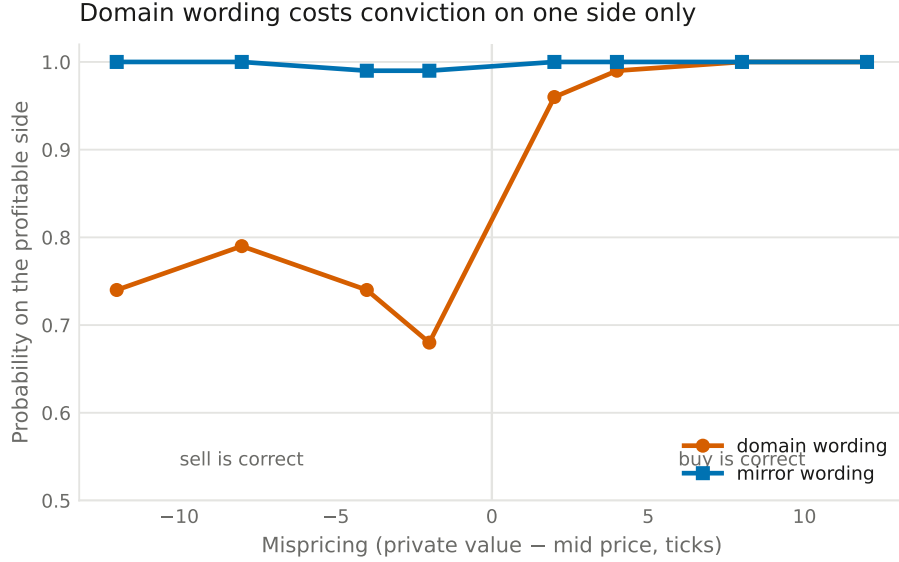


Figure 1: Probability the model places on the profitable side, against mispricing, under the two wordings. The model is symmetric when asked symmetrically. The effect is $25\times$ the model’s own run-to-run noise.

With $SE = 0.140$ the run had ample power to detect the exploratory effect, so this is evidence of absence.

We report it as such. Wording skews what the model says; it does not, in this market, change what prices do.

6 Experiment 2: liquidity failure under modal decoding

The null above was correct for the question it asked, and the question was wrong. Root mean squared pricing error is defined only over periods in which trade occurred, and so discards exactly the periods where the effect lives. In one run, modal decoding produced zero trades in all three up-jump windows and full trading in all three down-jump windows—270 trades in the run, none in the 45 periods after a positive shock.

We therefore registered a new outcome, defined for every period: $\text{traded}(d)$, the share of the fifteen periods after a jump of direction d in which at least one trade occurred.

Mechanism (exploratory diagnostic). We instrument resting depth on both sides at the close of every period, which separates two failures: *no counterparty* (one side empty) and *no crossing* (both sides present, prices do not meet). Table 2 classifies every silent post-shock period.

The failure is one-sided: the market runs out of *sellers*. It is essentially never short of buyers (0% of argmax silences, 4–7% of sampling silences). This classification was added after the effect was found and is exploratory corroboration, not an independent confirmatory test; E1 is the preregistered mechanism prediction.

Arm	Traded share		Gap
	after up	after down	
Jev-argmax	65.6%	91.3%	+25.8
Jev-sample	92.2%	95.3%	+3.1
ZI	76.0%	75.6%	−0.4
NBR	98.7%	99.7%	+1.0

Table 1: Preregistered confirmatory test, seeds 24–43. Levels and gaps are both computed from the same run; the gap column is the mean of per-seed gaps, which is why it need not equal the difference of the pooled levels exactly. D1: the Jev-argmax gap exceeds zero, $t = 6.37$, Holm $p < 10^{-5}$, eighteen of twenty seeds positive. D2: it exceeds the Jev-sample gap by 22.7 points, $t = 6.01$, Holm $p < 10^{-5}$.

Arm	Shock	No seller	No buyer	Both sides, no cross
Jev-argmax	up	100.0%	0.0%	0.0%
	down	96.2%	0.0%	3.8%
Jev-sample	up	71.4%	4.3%	24.3%
	down	57.1%	7.1%	35.7%
ZI	up	0.9%	2.8%	96.3%
	down	2.3%	1.8%	95.9%
NBR	up	8.3%	0.0%	91.7%
	down	0.0%	0.0%	100.0%

Table 2: Why each silent post-shock period was silent, twenty seeds. Under modal decoding the market is missing a counterparty essentially always; the symmetric baselines fall silent the ordinary way, with both sides quoting and prices failing to meet.

Unanimity is directional in the market. A static probe of the model, holding the book fixed at its pre-shock level, returns agreement of 92% after an up shock and 96% after a down shock. Taken at face value that inverts the result: more agreement on the side that halts less. We initially took this to mean unanimity could not account for the direction, and posited a second mechanism. That was wrong, and the error was using a probe statistic as though it were a market statistic. The probe holds the book frozen and sets the trader’s signal equal to its own private value; the market does neither.

Measured in the market, agreement is already directional (Table 3). In the first period after a shock, 90.0% of traders buy following an up shock, while after a down shock 59.4% sell against 34.6% buying. Unanimity after an up shock is near-total; after a down shock it is far weaker. That ordering matches the halt, and no second mechanism is required.

The effect therefore has a single mechanism. Unanimity is near-total after up shocks and materially weaker after down shocks, which sets both the magnitude of the halt and which shock it strikes; and it dissolves with market size exactly as Section 7 predicts. The sampling arm exhibits weaker correlation throughout, consistent with the size results.

What remains unexplained. Why the arm is more decisive on the buy side than the sell side at equal absolute mispricing—90.0% against 59.4% at the shock—is not established here. It

Period	After up shock			After down shock		
	Buy	Sell	Pass	Buy	Sell	Pass
t	90.0%	7.9%	2.1%	34.6%	59.4%	6.0%
$t + 1$	72.7%	23.1%	4.2%	51.0%	37.7%	11.2%
$t + 2$	72.1%	23.1%	4.8%	47.9%	40.4%	11.7%
$t + 3$	69.0%	25.8%	5.2%	47.5%	44.4%	8.1%
$t + 4$	67.5%	26.5%	6.0%	47.1%	45.0%	7.9%
window	65.9%	28.2%	5.8%	48.8%	42.3%	8.8%

Table 3: Jev-argmax order flow by period after a shock, twenty seeds. At the shock itself the arm is near-unanimous on the up side and takes the correct side on the down side. The sell majority is gone by $t + 1$, once the book has begun to move, so the fifteen-period window average understates what happens at the moment that matters. Reporting only the window would suggest the arm never reaches a sell majority, which is false.

is not reproduced by either symmetric baseline, and it is not removed by the mirror wording of Section 5, which eliminates the *stated* conviction asymmetry entirely. Candidate explanations we cannot separate with this design include the model itself, the option naming in the response schema, the under-specified referent of “the current market price” in the instructions, and the omission of inventory from the state. We flag it as the clearest target for follow-up.

A control that does not quite hold. NBR returns +1.0 points ($t = 3.33$), small but distinguishable from zero, indicating a slight genuine structural tilt in the market. It is 1/26 of the effect under test, ZI is clean (-0.4 , $t = -0.22$), and it does not survive the per-size controls of Section 7. We report it because the registration required controls to sit on zero and this one does not.

7 Experiment 3: the mechanism, tested where it could fail

If the market halts because every trader reaches the same conclusion, the effect must depend on the number of traders. With agreement rate p measured independently at 0.92, the probability that all N agree is approximately p^N , which falls from 0.72 at $N = 4$ to 0.07 at $N = 32$. We registered two predictions before running: that the argmax gap would decay with N (E1), and that the sampling gap would not (E2).

Statistics. The sweep reuses the same twenty seeds at every market size, so pooling all eighty observations into one regression would treat repeated measures as independent. We estimate a slope within each seed and test over seeds. We note also that the $N = 8$ argmax cell is the same data as D1.

E1 confirmed. The gap falls from 37.0 points at $N = 4$ to 4.2 at $N = 32$: slope -0.112 per doubling, seed-paired $t = -7.78$, nineteen of twenty seed slopes negative. (The pooled estimate, $t = -6.91$, understates it.) It is robust to dropping the $N = 4$ point, which we flagged in advance

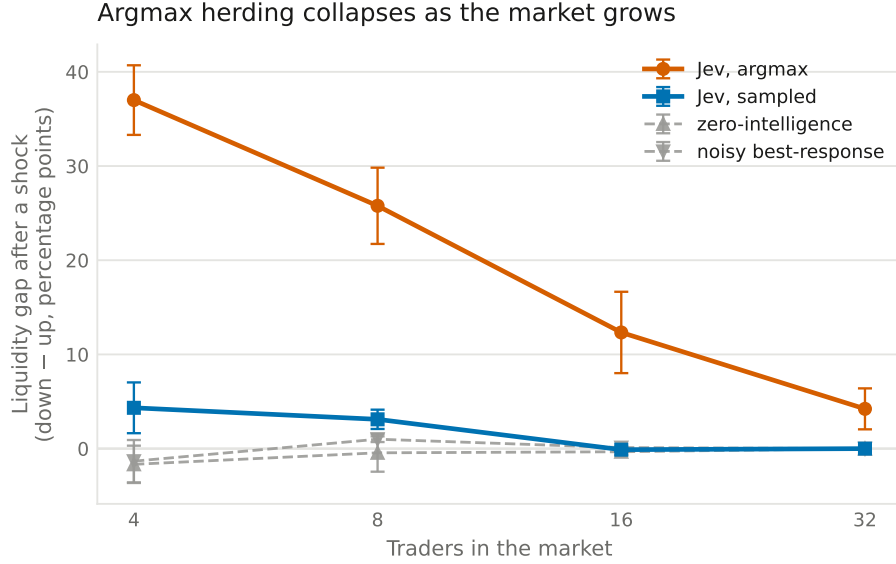


Figure 2: The liquidity gap against market size, twenty seeds per point, bars are standard errors. At $N = 32$ both symmetric baselines trade in 100% of periods with a zero gap, so there is no structural floor and the residual 4.2 points is attributable to the arm.

as floor-constrained because the symmetric baselines themselves trade in only 33% of periods there.

E2 not resolved. We predicted sampling would show no decay. The seed-paired estimate is -0.016 per doubling, $t = -2.00$, two-sided $p = 0.060$, with thirteen of twenty seed slopes negative. That does not falsify E2 at conventional levels. It also does not confirm it: no equivalence margin was preregistered, so a non-significant slope is not evidence of no slope. An earlier draft of this paper reported E2 as falsified on the pooled $t = -2.53$; that was an artefact of ignoring the repeated-measures structure, and we withdraw it.

What can be said is weaker and still useful. Sampling’s gap is small at every size (4.3, 3.1, -0.1 , 0.0 points) and its slope is roughly a seventh of the argmax slope, though the ratio rests on a marginally estimated denominator. Sampling does not abolish correlation; whether it decays with size on the same curve is unresolved here.

8 Discussion

The practical claim is narrow and, we think, general in shape: *the best way to use one model is not the best way to use many of them*. Modal decoding is individually optimal and collectively degenerate whenever agents share information and require counterparties or diversity. Sampling is individually worse—it deliberately takes a less likely action some of the time—and here it was the difference between a market that cleared and one that did not.

The failure is not a reasoning failure. The model’s modal action was correct on both sides at every mispricing tested. Nothing in its judgement was wrong; the loss occurred in aggregation. Evaluation of a single agent’s decisions cannot detect this, which is what makes it worth reporting.

9 Limitations

The origin of the directional bias. Section 6 establishes *that* the modal arm carries a standing buy bias and that this bias determines which shock the halt strikes. It does not establish *why*.

Two candidate explanations are not reproduced. It does not appear in the symmetric baselines at any market size, though a market feature could in principle produce an asymmetry for one policy and not another. And it is not removed by our question wording: under the mirror phrasing of Section 5, which removes the stated conviction asymmetry entirely (+0.005 against +0.250), the order-flow bias is unchanged—48.1% buys against 45.7% sells after a down shock, versus 48.8% against 42.3% under domain wording. Whatever produces the bias survives making the question symmetric.

What remains is the model itself, the task framing, or the state representation, and we cannot presently separate these. We flag it as the clearest target for follow-up work.

One model. All results are for a single typed-decision model. We have no evidence about others, and cannot say whether the agreement rate that drives the mechanism is characteristic or idiosyncratic.

Provenance of the probabilities. We treat the returned probabilities as the model’s beliefs. We have asked the vendor whether they are calibrated posterior mass or post-processed, and have no answer. If the latter, Experiment 1 is a statement about a serving stack rather than a model.

Full information only. Every trader observes F_t exactly, which maximises the correlation the mechanism exploits. The delayed and noisy treatments are implemented but unreported, and would plausibly weaken the effect.

Scale. Markets of four to thirty-two traders over 160 periods. The extrapolation to larger markets is the fitted decay, not a measurement.

10 Reproducibility

Because the model is non-deterministic and has no seed, code alone does not reproduce these numbers. Every response is archived—118,628 of them—and the analysis can be replayed against the archive with live calls disabled, in which case a cache miss raises rather than silently contacting the API. Total spend was approximately \$3.85 across roughly 118,628 live calls. These figures, and SHA-256 hashes of every result file and archive bundle, are generated from disk into `extttdata/manifest.json` rather than transcribed.

The preregistrations, including the falsified hypotheses and the reasons the outcome measure changed between Experiments 1 and 2, are in the repository with their commit timestamps.

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